EMBEDDED WRONG SIDE VEHICLE DETECTION AND INTELLIGENT TRAFFIC SIGNAL CONTROL USING ESP32

PROJECT-21ECP302L

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BONAFIDE CERTIFICATE

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ABSTRACT

This project focuses on the design and implementation of an intelligent traffic signal control system integrated with wrong-side vehicle detection using the ESP32 microcontroller. Wrong-side driving is a major cause of road accidents, especially in developing countries like India, where traffic violations frequently occur at intersections and one-way roads.

The proposed system utilizes infrared (IR) sensors to detect vehicle movement and determine direction based on the sequence of sensor activation. If a vehicle is detected moving in the wrong direction, the system generates an alert through a buzzer and warning indicator. Additionally, the ESP32 dynamically controls traffic signals (Red, Yellow, Green) based on traffic density, improving traffic flow efficiency.

Advanced features such as emergency vehicle priority and fail-safe signal operation enhance the reliability of the system. This project contributes to Sustainable Development Goal (SDG 11: Sustainable Cities and Communities) by promoting safer and smarter transportation systems.

ACKNOWLEDGEMENT

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ABBREVIATIONS

Abbreviation	Full Form
ESP32	Espressif 32-bit Microcontroller
IR	Infrared
LED	Light Emitting Diode
GPIO	General-Purpose Input/Output
IoT	Internet of Things
IDE	Integrated Development Environment
SDG	Sustainable Development Goal
PWM	Pulse Width Modulation
Wi-Fi	Wireless Fidelity
DC	Direct Current

CHAPTER 1

INTRODUCTION

1.1 Background and Motivation

With the rapid growth of urbanization and vehicular population, traffic management has become a critical challenge in modern cities. Road accidents and traffic congestion are increasing at an alarming rate, primarily due to traffic violations such as wrong-side driving. This issue is particularly severe in developing countries like India, where adherence to traffic rules is often inconsistent and monitoring systems are limited. [1-5]

Traditional traffic signal systems operate on fixed time intervals and lack the intelligence to adapt to real-time traffic conditions. These systems are incapable of detecting violations such as wrong-direction movement, leading to increased accident risks and inefficient traffic flow. The absence of automated monitoring mechanisms further aggravates the problem, as manual supervision is not always feasible at every junction. [6-10]

The motivation behind this project is to develop a low-cost, intelligent traffic management system using embedded technology that can detect wrong-side vehicle movement and dynamically control traffic signals. By integrating ESP32 with sensor-based detection, the system aims to enhance road safety, reduce congestion, and contribute to smart city development. [11-15]

1.2 Problem Statement

Despite advancements in traffic management technologies, many existing systems still suffer from significant limitations. Conventional traffic signal systems rely on fixed timing mechanisms, which do not adapt to changing traffic conditions. This results in inefficient traffic flow and increased waiting time at intersections.

Furthermore, most systems lack the capability to detect traffic violations such as wrong-side driving. Such violations often go unnoticed until accidents occur, posing serious risks to public safety. Existing intelligent systems that incorporate advanced technologies like computer vision or

IoT are often expensive and require complex infrastructure, making them unsuitable for widespread implementation.

Therefore, there is a need for a low-cost, efficient, and reliable traffic management system that can:

- Detect wrong-side vehicle movement in real time
- Automatically control traffic signals
- Adapt to traffic density
- Provide emergency vehicle priority
- Operate independently without complex infrastructure

1.3 Objectives of the Project

The primary objective of this project is to design and implement an intelligent traffic signal control system with wrong-side vehicle detection using the ESP32 microcontroller.

The specific objectives are as follows:

1. To design a low-cost embedded traffic management system using ESP32 and to detect vehicle movement and direction using infrared (IR) sensors.
2. To identify wrong-side vehicle movement based on sensor triggering sequence and to generate alerts using buzzer and indicators for violations.
3. To dynamically adjust signal timing based on traffic density and to provide priority for emergency vehicles using manual input.
4. To ensure fail-safe operation in case of system failure.

1.4 Importance of the Project

The importance of this project lies in its ability to address real-world traffic management challenges using a simple and cost-effective embedded system. Unlike conventional systems, the proposed solution combines violation detection with intelligent signal control, making it more efficient and practical.

This project demonstrates the application of embedded systems in solving societal problems such as road safety and traffic congestion. It also highlights the use of ESP32 as a powerful and versatile microcontroller for real-time applications.

Additionally, the system provides a scalable solution that can be extended for smart city infrastructure, automated monitoring, and IoT-based traffic management systems.

1.5 Sustainable Development Goal (SDG) Alignment

This project is aligned with **SDG 11: Sustainable Cities and Communities**, which focuses on developing safe, resilient, and sustainable urban environments.

By reducing traffic violations and improving traffic flow efficiency, the system contributes to safer roads and reduced accident rates. The use of low-cost embedded technology promotes sustainable infrastructure development, making the solution accessible and scalable.

Furthermore, the system supports smart city initiatives by integrating automation and real-time decision-making into traffic management.

1.6 Overview of the Report

This report is organized into five chapters:

- Chapter 1 – Introduction: Provides background, problem statement, objectives, and importance of the project.
- Chapter 2 – Literature Survey: Reviews existing traffic management systems and identifies research gaps.
- Chapter 3 – System Description and Methodology: Explains system design, hardware, software, and working methodology.
- Chapter 4 – Results and Discussion: Presents experimental results and system performance analysis.
- Chapter 5 – Conclusion: Summarizes the project outcomes and future scope.

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction to the Research Area

The rapid growth of urbanization and vehicular population has led to increasing challenges in traffic management systems across modern cities. Conventional traffic signal systems operate on fixed timing cycles, which are inefficient in handling dynamic traffic conditions and fail to address critical issues such as congestion, emergency vehicle delays, and traffic violations.

With advancements in embedded systems, Internet of Things (IoT), and computer vision technologies, intelligent traffic management systems have gained significant attention. These systems aim to optimize traffic flow, reduce waiting time, and enhance road safety through automation and real-time data processing.

Recent research focuses on integrating sensor-based detection, wireless communication, and artificial intelligence to create adaptive traffic control systems. However, despite these advancements, challenges such as system cost, infrastructure requirements, and lack of violation detection persist.

This chapter reviews key research works related to traffic density control, emergency vehicle prioritization, and intelligent traffic monitoring, forming the foundation for the proposed system.

2.2 Review of Existing Systems

Rohan Gupta et al. (2023) proposed an IoT-based traffic signal control system that dynamically adjusts traffic light timing based on real-time traffic density. The system utilizes sensors and real-time data analytics to monitor vehicle density at intersections. Based on the collected data, the traffic signal duration is automatically modified to improve traffic flow efficiency and reduce congestion. The study highlights that increasing traffic density leads to delays, fuel consumption, and environmental pollution, emphasizing the need for adaptive traffic control systems. The proposed system uses IoT-enabled devices, ultrasonic sensors, and servo motors to control traffic lights. It also integrates cameras to capture traffic violations and enhance monitoring capabilities. The proposed IoT-based traffic control system offers several advantages, including the ability to dynamically adjust signal timing based on real-time traffic density, which helps in optimizing traffic flow. This leads to a significant reduction in congestion and waiting time at intersections, thereby improving overall transportation efficiency. Additionally, the integration of IoT enables continuous real-time monitoring of traffic conditions, allowing better decision-making and control. [16]

Jyothsna Sunil et al. (2025) developed a computer vision-based intelligent traffic management system for ambulance routing using YOLO and IoT integration. The system uses cameras and deep learning algorithms to detect ambulances in real time. The ESP32 microcontroller processes data received from the camera and GPS module to dynamically control traffic signals and provide priority to emergency vehicles. The computer vision-based traffic management system using YOLO offers several advantages, particularly due to its use of deep learning techniques. It provides high accuracy in detecting ambulances, enabling reliable identification in real-time scenarios. This allows the system to prioritize emergency vehicles efficiently by dynamically controlling traffic signals. Additionally, the integration of IoT enables remote monitoring and control, making the system more flexible and suitable for smart city applications. [17]

Rajeshwari Sundar et al. (2015) proposed an intelligent traffic control system using RFID, GSM, and ZigBee technologies for congestion control, ambulance clearance, and stolen vehicle detection. In this system, each vehicle is equipped with an RFID tag, which is detected by RFID readers installed at traffic junctions. The system calculates traffic density based on the number of vehicles passing through a particular lane and dynamically adjusts the signal timing accordingly. The system also includes Emergency vehicle priority using ZigBee communication between ambulance and traffic signals and Stolen vehicle detection using RFID tag matching. [18]

2.3 Critical Observations from the Literature

From the analysis of the above research works, several important observations can be made:

- Most systems focus primarily on traffic density control, while ignoring violation detection such as wrong-side driving.
- Advanced systems using computer vision and AI provide high accuracy but are expensive and require significant computational resources.

- RFID-based systems offer automation but depend on vehicle-level hardware installation, which is not practical in all scenarios.
- IoT-based systems rely heavily on network connectivity, which may affect reliability in real-time applications.
- There is a lack of low-cost, embedded solutions that integrate both traffic control and violation detection.

2.4 Summary

The literature survey highlights the evolution of intelligent traffic management systems using IoT, RFID, and computer vision technologies. While existing systems effectively address issues such as traffic congestion and emergency vehicle prioritization, they fail to provide a comprehensive, low-cost solution that includes violation detection.

The proposed system builds upon these existing approaches by integrating wrong-side vehicle detection with intelligent traffic signal control using ESP32, offering a cost-effective, scalable, and efficient solution for modern traffic management.

CHAPTER 3

SYSTEM DESCRIPTION AND METHODOLOGY

3.1 Introduction

This chapter describes the overall design and methodology of the proposed intelligent traffic signal control system with wrong-side vehicle detection. The system is developed using the ESP32 microcontroller and integrates sensor-based detection, signal control logic, and safety mechanisms to improve traffic management.

The proposed system focuses on providing a low-cost, efficient, and real-time solution for traffic monitoring and violation detection. It combines hardware components such as IR sensors, LEDs, and buzzer with embedded software to achieve automation and reliability.

3.2 System Overview

The system consists of multiple components working together to monitor vehicle movement, detect violations, and control traffic signals. The ESP32 microcontroller acts as the central processing unit, which receives input from sensors and generates appropriate output signals.

The main functions of the system include:

- Detection of vehicle movement using IR sensors
- Identification of vehicle direction
- Detection of wrong-side driving
- Traffic signal control (Red, Yellow, Green)
- Traffic density-based signal timing
- Emergency vehicle prioritization
- Fail-safe operation in case of system failure

3.3 Block Diagram Description

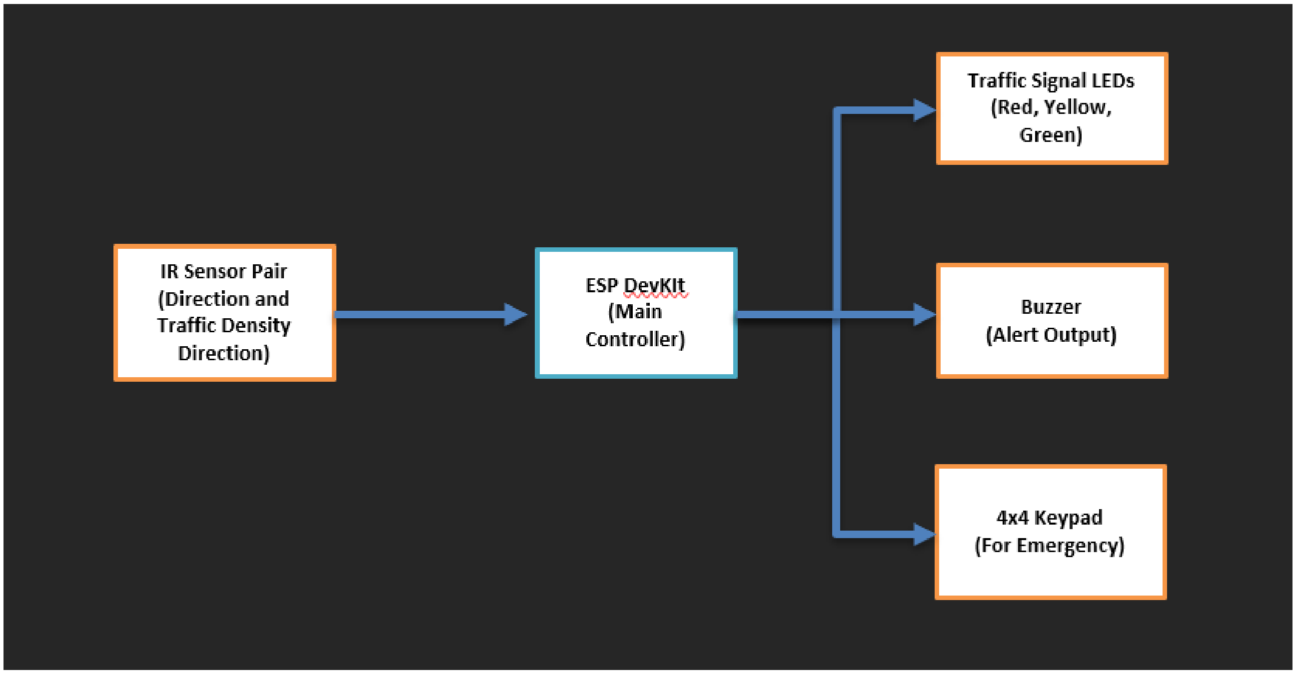

Figure 3.1: Block Diagram of Traffic Signal Control System Using ESP32

The block diagram of the proposed system is shown in Fig. 3.1, and it consists of the following components:

- **ESP32 Microcontroller:** Central controller of the system
- **IR Sensor Pair:** Used for vehicle detection and direction analysis
- **Traffic Signal LEDs:** Indicate signal status
- **Buzzer:** Generates alert during violations
- **Keypad:** Used for emergency vehicle input

The ESP32 receives signals from IR sensors and processes them to determine vehicle movement. Based on this analysis, it controls the traffic signals and activates alerts when necessary.

3.4 Circuit Design

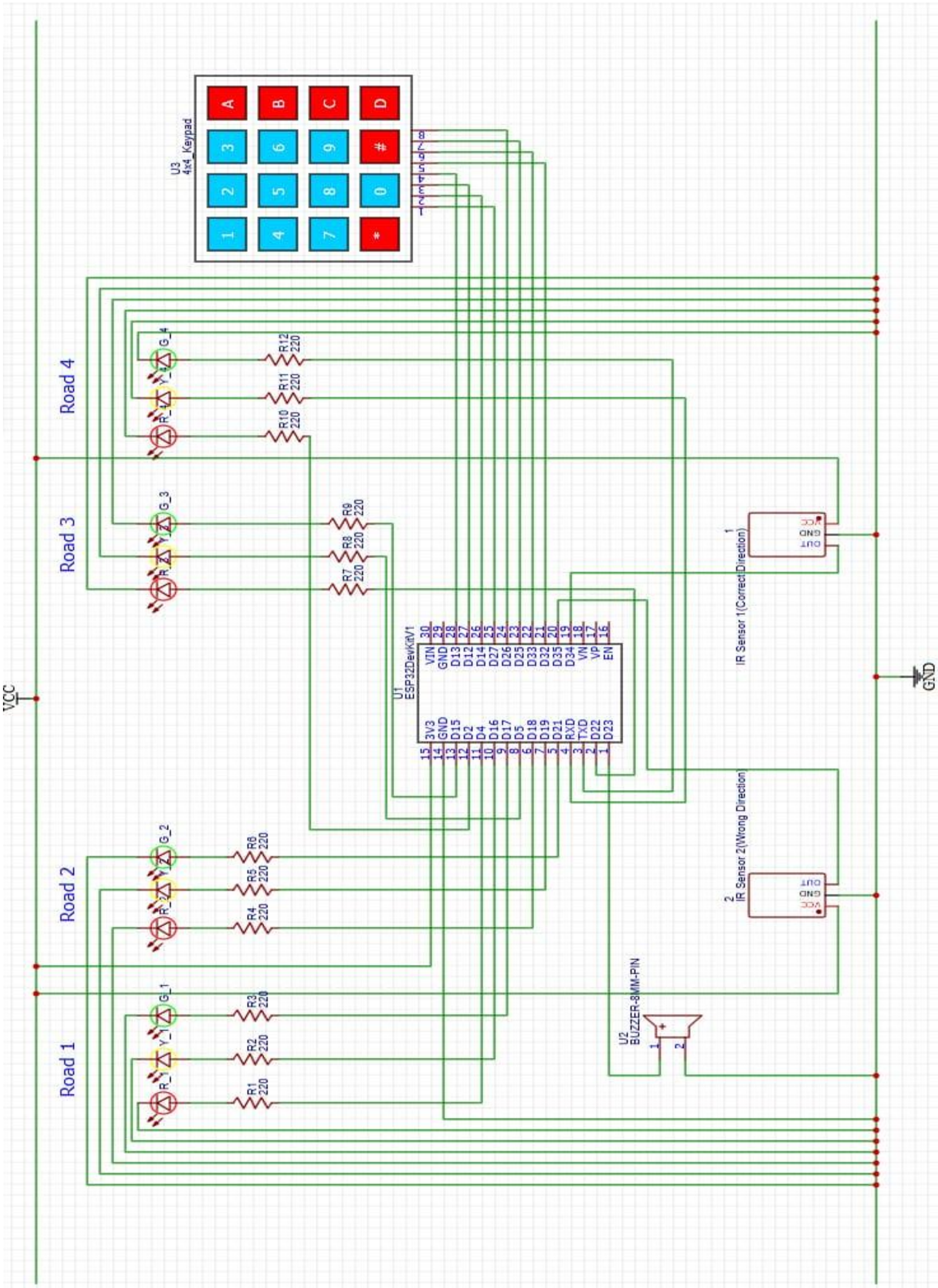

Figure 3.2: Hardware Circuit Design of Traffic Signal Control System Using ESP32

The circuit shown in Fig. 3.2 is designed using ESP32 GPIO pins connected to various components:

- IR sensors are connected as input devices
- LEDs are connected through resistors to represent traffic signals
- Buzzer is connected as output for alerts
- Keypad is interfaced for manual emergency input

3.5 Hardware Components

Table 3.1: Hardware Components Used in ESP32-Based Traffic Control System

Component Name	Quantity	Description / Purpose
ESP32 DevKit Microcontroller	1	Acts as the main control unit; processes sensor inputs and controls traffic signals and alert systems.
IR Sensor Modules	2	Used to detect vehicle presence and determine direction of movement based on sensor triggering sequence.
Traffic Signal LEDs (Red, Yellow, Green)	12 (3 per road × 4 roads)	Indicate traffic signal status for each lane (Red – Stop, Yellow – Wait, Green – Go).
Buzzer	1	Provides an audible alert when wrong-side vehicle movement is detected.
4x4 Matrix Keypad	1	Used to input emergency commands and provide priority signal to selected lanes.
Resistors (220Ω)	As required	Used to limit current for LEDs and ensure safe operation of components.
Connecting Wires	As required	Used for electrical connections between ESP32 and other components.
Breadboard	1	Used for assembling and testing the circuit without soldering.
Power Supply (5V Adapter/USB)	1	Provides regulated power to the ESP32 and connected hardware components.

3.6 Software Description

The system is programmed using Arduino IDE with Embedded C. The software controls:

- Sensor data processing
- Direction detection logic
- Signal timing control
- Emergency handling
- Alert generation

The program runs continuously in a loop, ensuring real-time system operation.

3.7 Working Methodology

The working of the system is divided into multiple steps:

Step 1: Vehicle Detection

IR sensors detect the presence of vehicles as they pass through the road.

Step 2: Direction Analysis

The system determines the direction of movement based on the sequence of sensor activation:

- Correct sequence → normal movement
- Reverse sequence → wrong-side movement

Step 3: Wrong-Side Detection

If a vehicle is detected moving in the wrong direction:

- Buzzer is activated
- Warning alert is generated
- Violation count is updated

Step 4: Traffic Signal Control

The ESP32 controls traffic signals in the following sequence:

- Green → Yellow → Red

Step 5: Traffic Density Detection

If a sensor remains blocked for a longer duration:

- The system interprets it as high traffic density
- Green signal duration is increased

Step 6: Emergency Vehicle Priority

When an emergency button is pressed:

- The corresponding lane receives green signal
- Other lanes are stopped

Step 7: Fail-Safe Operation

In case of system failure:

- The signal switches to blinking mode
- Ensures basic traffic safety

3.8 Algorithm

1. Start the system operation by powering the ESP32 microcontroller and initializing the program execution.
2. Initialize all input and output components, including IR sensors, traffic signal LEDs, buzzers, and keypad to ensure proper functioning.
3. Continuously read input signals from the IR sensors to detect the presence of vehicles on the road.
4. Analyze the sequence of sensor activation to determine the direction of vehicle movement (correct direction or wrong direction).
5. If a wrong-side vehicle is detected, immediately activate the buzzer to generate an alert and indicate a traffic violation.
6. If no violation is detected, allow the system to continue normal traffic signal operation without triggering any alerts.

7. Monitor the duration for which sensors remain blocked to estimate traffic density and adjust the green signal timing accordingly.
8. Check for emergency input from the keypad, and if detected, override the normal signal sequence to provide priority to the selected lane.
9. Update the traffic signals (Red, Yellow, Green) based on the current system logic, traffic conditions, and emergency inputs.
10. Repeat the entire process continuously to ensure real-time monitoring, detection, and control of traffic flow.

3.9 Advantages of the Proposed System

- Low-cost implementation
- Real-time detection and response
- Improved traffic efficiency
- Enhanced road safety
- Scalable for smart city applications

CHAPTER 4

RESULTS AND DISCUSSION

This chapter presents the experimental results and performance analysis of the proposed intelligent traffic signal control system with wrong-side vehicle detection. The system was implemented using ESP32 and tested under different traffic scenarios to evaluate its effectiveness, accuracy, and response time.

The results demonstrate the system's ability to detect vehicle direction, control traffic signals dynamically, and respond to emergency situations efficiently.

4.1 Normal Traffic Operation

In normal conditions, when vehicles move in the correct direction, the system operates as a standard traffic signal controller. The infrared sensors detect vehicle presence, and the ESP32 processes the signals to maintain the regular traffic light sequence.

The system follows the sequence:
Green → Yellow → Red, ensuring smooth and orderly traffic flow. No alerts are generated during this condition, indicating proper functioning of the system. This verifies that the system does not produce false positives and maintains stability during normal operation.

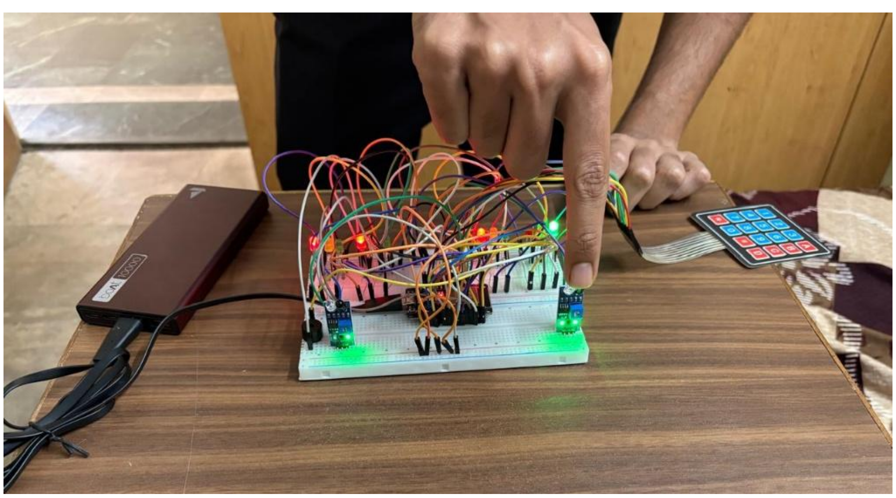

Figure 4.1: Normal Traffic Signal Operation

The Fig. 4.1 illustrates the normal operation of the traffic signal system, where vehicles move in the correct direction and the ESP32 controls the signal sequence in the order of Green → Yellow → Red. No alerts or violations are detected, indicating stable and correct system functioning.

4.2 Wrong-Side Vehicle Detection

When a vehicle moves in the wrong direction, the sequence of IR sensor activation is reversed. The ESP32 detects this abnormal pattern and identifies it as a violation.

Upon detection:

- The buzzer is immediately activated
- An alert is generated
- The system records the violation

This demonstrates the system's capability to detect wrong-side movement in real time and respond instantly, improving road safety.

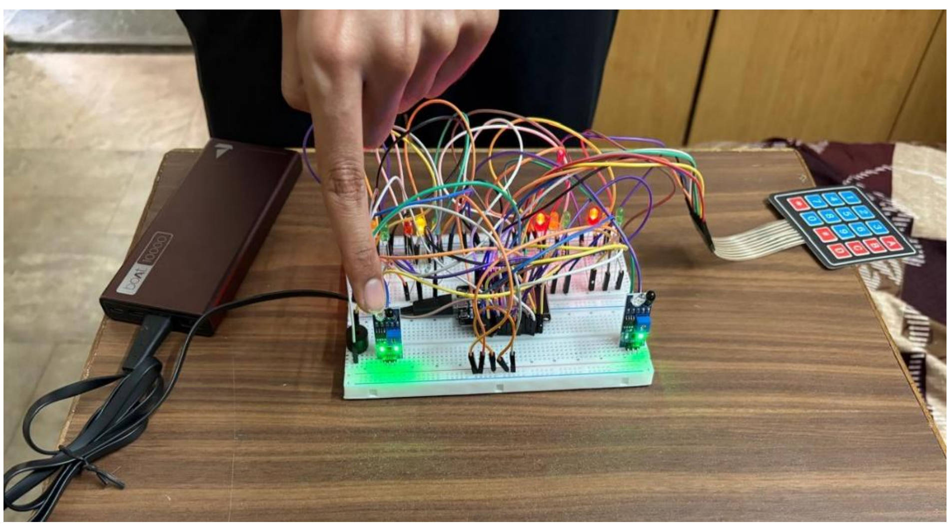

Figure 4.2: Wrong-Side Vehicle Detection and Alert Activation

The Fig. 4.2 shows the detection of a wrong-side vehicle movement based on reversed IR sensor activation. Upon identifying the violation, the system activates a buzzer and generates an alert, demonstrating real-time violation detection and response.

4.3 Traffic Density-Based Signal Control

The system dynamically adjusts traffic signal timing based on vehicle density. When a sensor remains blocked for a longer duration, it indicates high traffic density.

In such cases:

- The green signal duration is increased
- More vehicles are allowed to pass
- Congestion is reduced

This adaptive behaviour improves traffic flow efficiency compared to fixed-time signal systems.

4.4 Emergency Vehicle Priority

The system includes a manual emergency override using a keypad. When an emergency input is provided, the system immediately prioritizes the selected lane.

During testing:

- The selected lane received a green signal
- Other lanes were stopped
- Emergency passage was ensured

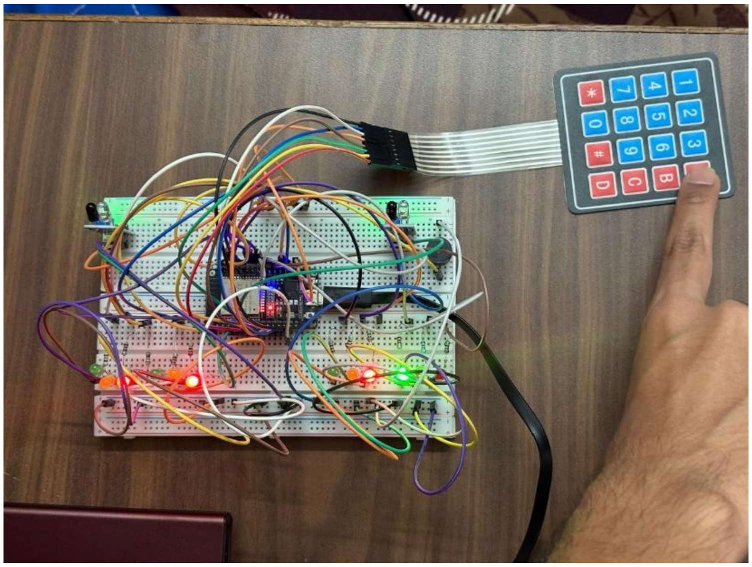

Figure 4.3: Emergency vehicle priority activated for Lane-A

The Fig. 4.3 represents the activation of emergency priority for Lane-A, where the system overrides normal operation and provides a green signal to Lane-A while all other lanes remain in the red state to ensure safe and quick passage.

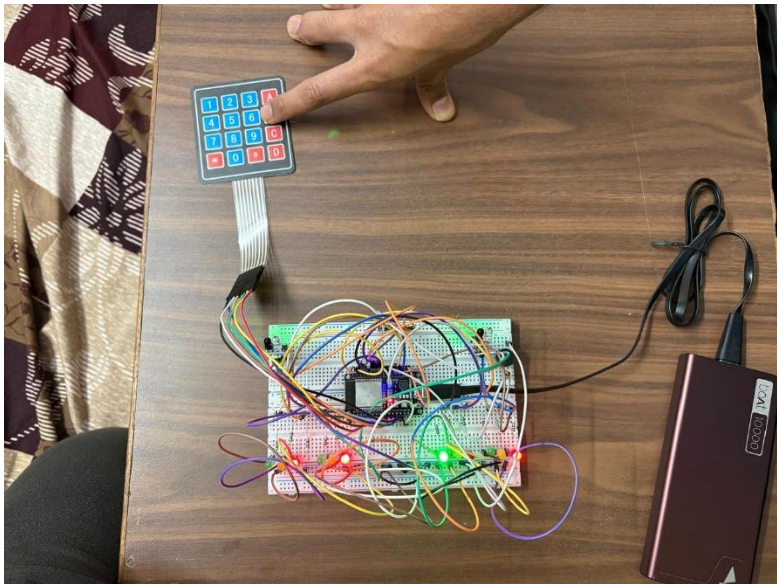

Figure 4.4: Emergency vehicle priority activated for Lane-B

The Fig. 4.4 depicts emergency vehicle prioritization for Lane-B. The system dynamically assigns a green signal to Lane-B and halts traffic in all other lanes, ensuring uninterrupted movement for emergency vehicles.

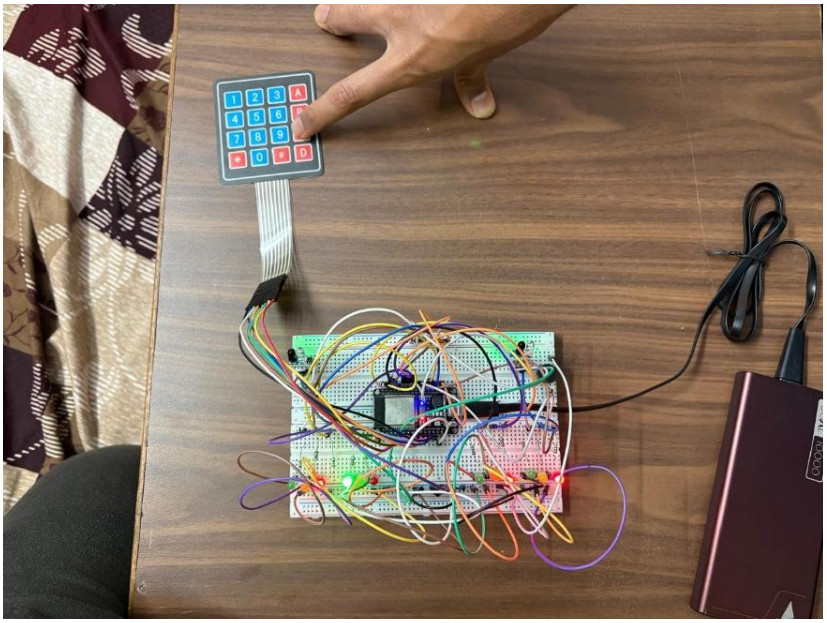

Figure 4.5: Emergency vehicle priority activated for Lane-C

The Fig. 4.5 shows the system providing emergency priority to Lane-C. The selected lane is given a green signal, while other lanes are stopped, demonstrating the system's ability to handle real-time emergency scenarios efficiently.

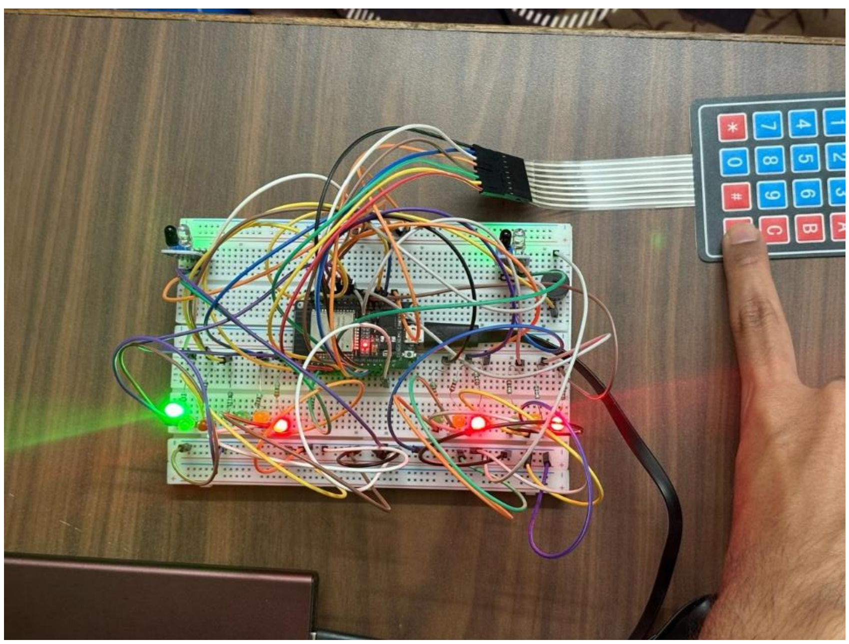

Figure 4.6: Emergency vehicle priority activated for Lane-D

The Fig. 4.6 illustrates emergency vehicle prioritization for Lane-D, where the system grants immediate green signal access to Lane-D and restricts all other lanes, ensuring quick and safe clearance.

4.5 Experimental Results Table

Table 4.1: System Performance Under Different Conditions

Test Condition	Expected Output	Observed Output	Result
Normal Traffic Movement	No alert, normal signal operation	System followed correct signal sequence	Successful
Wrong Direction Movement	Buzzer alert activation	Buzzer activated immediately	Successful
High Traffic Density	Increased green signal time	Green time extended	Successful
Emergency Input Activated	Priority green signal	Lane received immediate green	Successful
Continuous Operation	Stable system behaviour	No failure observed	Successful

4.6 Performance Analysis

The system demonstrated reliable performance under all tested conditions. It successfully detected wrong-side vehicle movement with high accuracy and responded instantly by activating alerts. The traffic density detection mechanism effectively adjusted signal timing, reducing congestion and improving flow.

The emergency priority feature functioned as expected, ensuring immediate clearance for selected lanes. The overall system response was fast, with minimal delay between detection and action.

4.8 Discussion

The results indicate that the proposed system is effective in addressing key traffic management challenges. Unlike conventional systems, it provides intelligent control and real-time monitoring.

However, certain limitations were observed:

- IR sensors may be affected by environmental conditions
- System is prototype-level
- No camera-based validation

Despite these limitations, the system offers a strong foundation for further development and real-world implementation.

CHAPTER 5

CONCLUSION

5.1 Conclusion

The project successfully demonstrates the design and implementation of an intelligent traffic signal control system integrated with wrong-side vehicle detection using the ESP32 microcontroller. The system effectively detects vehicle movement, identifies violations, and controls traffic signals dynamically based on real-time conditions.

The use of infrared sensors enables accurate detection of vehicle direction, while the embedded logic ensures timely response to violations. The system also improves traffic efficiency through adaptive signal timing and provides emergency vehicle priority, making it a comprehensive solution for modern traffic management.

Overall, the proposed system achieves its objectives of enhancing road safety, reducing congestion, and providing a low-cost, scalable solution suitable for smart city applications.

5.2 Future Scope

The proposed system can be further enhanced in several ways to improve performance and applicability:

- Integration of cameras for visual verification
- Use of AI/ML for advanced traffic analysis
- Cloud-based monitoring and data storage
- Mobile application for traffic control
- Integration with smart city infrastructure

5.3 Sustainable Development Goals (SDG) Mapping

Table 5.1: SDG Alignment of the Proposed System

SDG Goal	Description	Contribution of Project
SDG 11	Sustainable Cities and Communities	Improves urban traffic management and reduces accidents

5.4 Final Remarks

The project highlights the importance of integrating embedded systems and intelligent algorithms in traffic management. The system has the potential to significantly contribute to safer and smarter transportation system with further improvements and real-world deployment.

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